Jake Welde, Ph.D.

Assistant Professor
Sibley School of Mechanical and Aerospace Engineering
Cornell University

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My work sits at the intersection of robotics, control theory, differential geometry, and design. I am broadly interested in how mathematical structure—symmetry, hierarchy, and mechanics—can be harnessed to create more capable algorithms for robot control, and how the interplay between control and morphology mediates overall robot capabilities. At Cornell, I lead the Geometry, Design, and Control Laboratory—the "GeoDesiC Lab"↗. In our research, we seek answers to fundamental questions in control theory and dynamical systems, and we build robots that strive to draw nearer to the incredible example of Nature.

EMPLOYMENT

Assistant Professor, Sibley School of Mechanical and Aerospace Engineering 2025 - present
Cornell University *Ithaca, NY*
Principal Investigator of the Geometry, Design, and Control Laboratory ("GeoDesiC Lab") ↗ and member of the fields of Mechanical Engineering, Aerospace Engineering, Theoretical and Applied Mechanics, and Applied Mathematics.

EDUCATION

Doctor of Philosophy, Mechanical Engineering and Applied Mechanics 2025
University of Pennsylvania, General Robotics, Automation, Sensing, and Perception (GRASP) Laboratory *Philadelphia, PA*
Advisor: Vijay Kumar **Thesis Committee**: Daniel Koditschek, Michael Posa, Muruhan Rathinam, and Jim Ostrowski
Thesis: "Geometric Methods for Efficient and Explainable Control of Underactuated Robotic Systems"↗
Certificate in College and University Teaching, *Center for Excellence in Teaching, Learning and Innovation (CETLI).*

Master of Science in Engineering, Robotics 2020
University of Pennsylvania *Philadelphia, PA*

Bachelor of Science in Engineering, Mechanical Engineering and Applied Mechanics, *magna cum laude* 2019
University of Pennsylvania *Philadelphia, PA*
Minor in French and Francophone Studies. *Activities:* Underground Shakespeare Company

HONORS AND AWARDS

Best Poster (presented by my PhD student), *IROS Workshop on Geometric Representations in Robotics* 2026

RSS Pioneers Cohort Member, *Robotics: Science and Systems (RSS)* 2024
1 of 30 of the "world's top early-career researchers" in robotics, selected for an intensive career workshop (15% acceptance rate).

John A. Goff Prize, Mechanical Engineering and Applied Mechanics, *University of Pennsylvania* 2024
"Awarded annually to a graduate student... selected by the faculty on the basis of criteria of scholarship, resourcefulness, and leadership" (1-2 PhD students per year).

Best Contribution (Neuroscience & Interpretability Track), *NeurReps Workshop at NeurIPS* 2024

Best Paper Finalist (Unmanned Aerial Vehicles), *IEEE International Conference on Robotics and Automation (ICRA)* 2021

Outstanding TA Award, Mechanical Engineering and Applied Mechanics, *University of Pennsylvania* 2021

National Science Foundation Graduate Research Fellowship (NSF GRFP) 2019

PUBLICATIONS

PREPRINTS

1. "**A Weak Notion of Symmetry for Control Systems**",
   Jake Welde, Riley Link, and Pieter van Goor.
   Preprint, 2026.

JOURNAL ARTICLES

1. "**Almost Global Asymptotic Trajectory Tracking for Fully-Actuated Mechanical Systems on Homogeneous Riemannian Manifolds**",
   Jake Welde and Vijay Kumar.
   IEEE Control Systems Letters (L-CSS), 2024.
2. "**A Compositional Approach to Certifying Almost Global Asymptotic Stability of Cascade Systems**",
   Jake Welde, Matthew D. Kvalheim, and Vijay Kumar.
   IEEE Control Systems Letters (L-CSS), 2023.
3. "**Dynamically Feasible Task Space Planning for Underactuated Aerial Manipulators**",
   Jake Welde, James Paulos, and Vijay Kumar.
   IEEE Robotics and Automation Letters (RA-L), 2021. **Best Paper Finalist (Unmanned Aerial Vehicles) at ICRA 2021.**
4. "**Autonomous Flight for Detection, Localization, and Tracking of Moving Targets With a Small Quadrotor**",
   Justin Thomas, **Jake Welde**, Giuseppe Loianno, Kostas Daniilidis, and Vijay Kumar.
   IEEE Robotics and Automation Letters (RA-L), 2017.

REFEREED CONFERENCE PROCEEDINGS

1. "**A Weak Notion of Symmetry for Dynamical Systems**",
   Jake Welde and Pieter van Goor.
   IEEE Conference on Decision and Control (CDC), 2026.
2. "**Scalable Distributed Nonlinear Control Under Flatness-Preserving Coupling**",
   Fengjun Yang, **Jake Welde**, and Nikolai Matni.
   American Control Conference (ACC), 2026.
3. "**Learning Flatness-Preserving Residuals for Pure-Feedback Systems**",
   Fengjun Yang, **Jake Welde**, and Nikolai Matni.
   IEEE Conference on Decision and Control (CDC), 2025.
4. "**Leveling the Playing Field: Carefully Comparing Classical and Learned Controllers for Quadrotor Trajectory Tracking**",
   Pratik Kunapuli, **Jake Welde**, Dinesh Jayaraman, and Vijay Kumar.
   Robotics: Science and Systems (RSS), 2025.
5. "**Leveraging Symmetry to Accelerate Learning of Trajectory Tracking Controllers for Free-Flying Robotic Systems**",
   Jake Welde*, Nishanth Rao*, Pratik Kunapuli*, Dinesh Jayaraman, and Vijay Kumar.
   IEEE International Conference on Robotics and Automation (ICRA), 2025. **equal contribution.*
6. "**The Role of Symmetry in Constructing Geometric Flat Outputs for Free-Flying Robotic Systems**",
   Jake Welde, Matthew D. Kvalheim, and Vijay Kumar.
   IEEE International Conference on Robotics and Automation (ICRA), 2023.
7. "**Trajectory Planning for the Bidirectional Quadrotor as a Differentially Flat Hybrid System**",
   Katherine Mao, **Jake Welde**, M. Ani Hsieh, and Vijay Kumar.
   IEEE International Conference on Robotics and Automation (ICRA), 2023.
8. "**Coordinate-Free Dynamics and Differential Flatness of a Class of 6DOF Aerial Manipulators**",
   Jake Welde and Vijay Kumar.
   IEEE International Conference on Robotics and Automation (ICRA), 2020.

WORKSHOP PAPERS AND POSTERS *(NON-ARCHIVAL)*

1. "**Reduction of Tracking Control Problems under Weak Symmetry**",
   Riley Link, Pieter van Goor, and **Jake Welde**.
   Workshop on Geometric Representations in Robotics, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. **Best Poster.**
2. "**Learning Flatness-Preserving Residuals for Pure-Feedback Systems**",
   Fengjun Yang, **Jake Welde**, and Nikolai Matni.
   Workshop on Structured Learning, IEEE International Conference on Robotics and Automation (ICRA), 2025.
3. "**Geometric Methods of Systematic Controller Synthesis for Underactuated Robotic Systems**",
   Jake Welde.
   RSS Pioneers Workshop, Robotics: Science and Systems (RSS), 2024.
4. "**Integrated Hardware and Software Codesign for Controlling Underactuated Aerial Robots**",
   Jack Campanella, **Jake Welde**, and Vijay Kumar.
   Northeast Systems and Control Workshop (NESCW), 2024.
5. "**Leveraging Symmetry to Accelerate Learning of Trajectory Tracking Controllers for Free-Flying Robotic Systems**",
   Jake Welde*, Nishanth Rao*, Pratik Kunapuli*, Dinesh Jayaraman, and Vijay Kumar.
   Workshop on Symmetry and Geometry in Neural Representations, Conference on Neural Information Processing Systems (NeurIPS), 2024. **Best Contribution (Neuroscience & Interpretability Track).** **equal contribution.*
6. "**Towards a Lightweight Fully-Actuated Aerial Vehicle: Thrust Vectoring and Control Allocation Under Redundancy**",
   Saibernard Yogendran, **Jake Welde**, and Vijay Kumar.
   Northeast Systems and Control Workshop (NESCW), 2024.
7. "**Towards Automatic Identification of Globally Valid Geometric Flat Outputs via Numerical Optimization**",
   Jake Welde and Vijay Kumar.
   Workshop on Geometric Representations, IEEE International Conference on Robotics and Automation (ICRA), 2023.
8. "**Some Aerial Manipulators Can Exactly Track Arbitrary Smooth End-Effector Trajectories in 6 Degrees of Freedom**",
   Jake Welde and Vijay Kumar.
   Northeast Robotics Colloquium (NERC), 2019.

TALKS

INVITED TALKS

- **Syracuse University**, Syracuse, NY — 2027
  Department of Mechanical and Aerospace Engineering (forthcoming)
- **IEEE Conference on Decision and Control (CDC)**, Honolulu, HI — 2026
  Invited Session on "New Directions in Geometric Control Theory" (forthcoming)
- **Society of Engineering Science Annual Technical Meeting**, West Lafayette, IN — 2026
  Eringen Medal Symposium in Honor of Vijay Kumar
- **IAS Frontiers Conference on Geometry, Dynamics, and Learning (GDL)**, Singapore — 2026
  Nanyang Technological University
- **University of Sydney**, Sydney, Australia — 2026
  Australian Centre for Robotics
- **Robotics: Science and Systems (RSS)**, Sydney, Australia — 2026
  Workshop on "The Geometry of Motion: Physics-Informed Structures for Learning and Control"
- **Cornell University**, Ithaca, NY — 2025
  Center for Applied Mathematics

- **Cornell University**, Ithaca, NY 2025
  Robotics Seminar, Department of Computer Science
- **Massachusetts Institute of Technology**, Cambridge, MA 2025
  Department of Mechanical Engineering
- **Massachusetts Institute of Technology**, Cambridge, MA 2025
  Department of Electrical Engineering and Computer Science
- **Colorado State University**, Fort Collins, CO 2025
  Department of Mechanical Engineering
- **University of Minnesota**, Minneapolis, MN 2025
  Department of Aerospace Engineering and Mechanics
- **North Carolina State University**, Raleigh, NC 2025
  Department of Electrical and Computer Engineering
- **Cornell University**, Ithaca, NY 2025
  Sibley School of Mechanical and Aerospace Engineering
- **Boston University**, Boston, MA 2025
  Department of Mechanical Engineering
- **Villanova University**, Villanova, PA 2025
  Department of Mechanical Engineering
- **Arizona State University**, Mesa, AZ 2025
  School of Manufacturing Systems and Networks
- **University of Texas at Dallas**, Dallas, TX 2025
  Department of Systems Engineering
- **Joint Mathematics Meetings**, San Francisco, CA 2024
  Special Session on "Geometry and Symmetry in Differential Equations, Control, and Applications"

OTHER TALKS

- **Joint Mathematics Meetings**, Seattle, WA 2025
  Special Session on "Nonlinear Algebraic Methods in Artificial Intelligence and Machine Learning"
- **Northeast Systems and Control Workshop**, Philadelphia, PA 2024
  University of Pennsylvania
- **International Conference on Robotics and Automation (ICRA)**, London, UK 2023
  Workshop on "Compositional Robotics: Mathematics and Tools"
- **APS March Meeting**, Las Vegas, NV 2023
  Robophysics Focus Session
- **SIAM Conference on Control and Its Applications**, Philadelphia, PA 2023
  Contributed Lecture

MEDIA

"Exploring the Interplay of Math and Machines"↗, *Cornell Duffield Engineering* 2025

"Robots in the Reading Room: GRASP Lab Brings Hands-On STEM to Roxborough Library"↗, *Penn Engineering* 2025

"MEAM 520 Class Breakdown"↗, *GRASP Laboratory* 2022

"Virtual Robots: Taking Risks in an Online Classroom"↗, *Penn Engineering* 2021

"Game of Drones"↗, *National Geographic's "Breakthrough"* 2017

"Penn Students Create Gingerbread Replica of Fisher Fine Arts Library"↗, *34th Street Magazine* 2017

TEACHING

- **MAE 2030: Dynamics**, Cornell University — Fall 2025 / Fall 2026
  Instructor
- **MEAM 520: Introduction to Robotics**, University of Pennsylvania — Fall 2020 / Fall 2021
  Head Teaching Assistant for Prof. Cynthia Sung and Prof. M. Ani Hsieh
- **CIT 520: Introduction to Robotics**, University of Pennsylvania (online) — Spring 2021
  Teaching Assistant for Prof. Vijay Kumar
- **MEAM 211: Engineering Mechanics - Dynamics**, University of Pennsylvania — Spring 2021
  Head Teaching Assistant for Prof. Michael Posa

RESEARCH SUPERVISION *(AT CORNELL)*

PHD STUDENTS

- **Riley Link**, Applied Mathematics — 2026 - present
- **Daniel Whitman**, Robotics — 2026 - present
- **Ekaterina (Kat) Skorniakova**, Robotics — 2026 - present

MASTER OF ENGINEERING STUDENTS

- **Sotiris Batategas**, Aerospace Engineering — 2026 - present
- **Ava Ianuale**, Mechanical Engineering — 2026 - present
- **Ben Roitman**, Aerospace Engineering — 2026 - present
- **Delaney Lopez**, Mechanical Engineering — 2026 - present
- **Anthony Parlatore**, Mechanical Engineering — 2026 - present

UNDERGRADUATE STUDENTS

- **Utku Melemetci**, Computer Science — 2026 - present
- **Joseph Cobb**, Electrical and Computer Engineering — 2026 - present
- **Audrey Ho**, Mechanical Engineering — 2025 - present
- **Ollie Aizer**, Mechanical Engineering — 2025 - present
- **Lindsay Kossoff**, Mechanical Engineering — 2025 - present
- **Ronan Alo**, Mechanical Engineering — 2025 - present
- **Jonathan Song**, Electrical and Computer Engineering — 2025 - 2026
- **Andre Boufama**, Mechanical Engineering — 2025 - 2026

RESEARCH MENTORSHIP *(PRIOR TO CORNELL)*

PHD STUDENTS

- **Pratik Kunapuli**, Computer and Information Science, University of Pennsylvania (now at Eka Robotics) — 2024
- **Katie Mao**, Mechanical Engineering and Applied Mechanics, University of Pennsylvania — 2022

MASTER OF SCIENCE STUDENTS

- **Nishanth Rao**, Robotics (with thesis), University of Pennsylvania (now a PhD student at Princeton) — 2024 - 2025
- **Jack Campanella**, Robotics (with thesis), University of Pennsylvania (now at Vertiq) — 2023 - 2025
- **Saibernard Yogendran**, Robotics, University of Pennsylvania (now at ASML) — 2022 - 2024

UNDERGRADUATE STUDENTS

- **Eshan Singhal**, Computer Engineering and Physics, University of Pennsylvania — 2024 - 2025
- **Nicole Luna**, Mechanical Engineering and Physics, Cal Poly Pomona (now a PhD student at CU Boulder) — 2021

- **Natasha Dilamani**, Mechanical Engineering, University of Pennsylvania (now at Millennium) 2020

SERVICE

EDITORIAL AND PROFESSIONAL SERVICE

- **Associate Editor**, *IEEE Robotics and Automation Letters (RA-L)* 2026 - present
- **Area Chair**, *Robotics: Science and Systems (RSS)* 2026, 2027
- **Associate Editor** (Theoretical Foundations), *IEEE International Conference on Robotics and Automation (ICRA)* 2027
- **Associate Editor**, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* 2026
- **Member**, *Technical Committee on Nonlinear Systems and Control*, IEEE Control Systems Society 2026 - present

WORKSHOP ORGANIZATION

- **"Geometry in the Age of Data-Driven Robotics"** 2026
  IEEE International Conference on Robotics and Automation (ICRA)
  Organizers: Riddhiman Laha, Tobias Löw, and Jake Welde
- **"Equivariant Systems: Theory and Applications in State Estimation, Artificial Intelligence and Control"** 2025
  Robotics: Science and Systems (RSS)
  Organizers: Stephan Weiss, Jake Welde, Maani Ghaffari, and Robert Mahony
- **"RSS Pioneers"** (Early-Career Workshop) 2025
  Robotics: Science and Systems (RSS)
  Organizers: link↗ **Faculty Chairs**: Jake Welde, Frederike Dümbgen, and Yunlong Song
- **"Equivariant Robotics: The Role of Symmetry Across Perception, Estimation, and Control"** 2024
  IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
  Organizers: Jake Welde, Pieter van Goor, Yinshuang Xu, Rui Wang, Evangelos Chatzipantazis, Christine Allen-Blanchette, Kostas Daniilidis, and Vijay Kumar

REVIEW ACTIVITIES

- International Journal of Robotics Research (IJRR)
- IFAC Automatica
- IEEE/ASME Transactions on Mechatronics (TMECH)
- IEEE Control Systems Letters (L-CSS)
- Springer Autonomous Robots
- IEEE Robotics and Automation Letters (RA-L)
- ASME Journal of Dynamic Systems, Measurement and Control
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE Transactions on Robotics (T-RO)
- Robotics: Science and Systems (RSS)
- IEEE Transactions on Automatic Control (TAC)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE International Conference on Automation Science and Engineering (CASE)

DISSERTATION COMMITTEES

- **Jovan Menezes**, Mechanical Engineering, Cornell University
- **Chuwei Wang**, Mechanical Engineering, Cornell University
- **Noa Kaplan**, Computer Science, Cornell University
- **Fengjun Yang**, Computer and Information Science, University of Pennsylvania

- **Pratik Kunapuli**, Computer and Information Science, University of Pennsylvania

OUTREACH

- **Cornell University Physical Intelligence (CUPI)**, Faculty Advisor 2025 - present
- **Science Olympiad at the University of Pennsylvania**, Event Supervisor and Placement Leader 2017 - 2025
- **"Fun with Robots" Outreach Programming**, Roxborough Library, Free Library of Philadelphia 2024

FUN FACTS

My Erdős number↗ is at most 3 (Jake Welde ↔ Kostas Daniilidis ↔ Pavel Valtr ↔ Paul Erdős).

In my free time, I love spending time in Nature, cooking spicy food, and playing fetch with my very energetic dog Sprout.